

Disconnected Landlords — PI-Feedback Task Exhibits

Admin Data pipeline (branch `Full-Sample`)

2026-07-09 (LA geography)

This document assembles the tables and figures for the four PI-feedback tasks. **Sign conventions:** `current_energy_efficiency` (EPC/SAP score) is *higher = more efficient (better)*; matched-pair effects are treated minus control, so a *positive* value means the ownership type is *more efficient than its matched private-individual control*. `bad_epc_c` is the share below EPC band C (*higher = worse*). Archetypes are referred to throughout by the short names defined in Table 1.

Contents

1	Heterogeneous Treatment Effects and Property Archetypes	2
2	Matching Attrition	6
3	Transaction-Vintage Gradient	7
4	Foreign Non-Profits and Tax Havens	11

1 Heterogeneous Treatment Effects and Property Archetypes

Task 1. Matched-pair CATEs by property archetype, composition vs. like-for-like reweighting, the distinctive treated stock per ownership type, and the interpretable archetype $\times$ ownership matrix with baseline anchors. Table 1 is the archetype legend: the short name and the covariates that define each archetype.

Table 1: Property Archetypes: Short Names and Defining Covariates

	Short name	Property type	Built form	Age band	Main fuel	Floor tercile	Baseline EPC
A1	Edwardian terrace (medium)	House	Mid-Terrace	1900-1929	Mains gas	medium	62.1
A2	Edwardian flat	Flat	Mid-Terrace	1900-1929	Mains gas	small	67.0
A3	Post-war semi	House	Semi-Detached	1950-1966	Mains gas	medium	63.5
A4	Edwardian electric flat	Flat	Mid-Terrace	1900-1929	Electricity	small	56.2
A5	1980s terrace	House	Mid-Terrace	1983-1990	Mains gas	small	69.9
A6	Victorian oil detached	House	Detached	before 1900	Oil	large	42.7
A7	Post-war bungalow	Bungalow	Semi-Detached	1950-1966	Mains gas	small	63.1
A8	Edwardian terrace (large)	House	Mid-Terrace	1900-1929	Mains gas	large	60.9
A9	Interwar semi	House	Semi-Detached	1930-1949	Mains gas	large	60.6
A10	Victorian terrace	House	Mid-Terrace	before 1900	Mains gas	medium	61.1
<i>Council-stock reference (representative social housing)</i>							
R	Council flat	Flat	Semi-Detached	1950-1966	Mains gas	small	66.7
R	Council semi	House	Semi-Detached	1950-1966	Mains gas	medium	63.5

Notes: Each archetype is a covariate cell = property type $\times$ built form $\times$ construction age band $\times$ main fuel $\times$ floor-area tercile, in the eligible private-rental control pool, England-wide. Archetypes (A) are a diversity-aware selection spanning property type $\times$ fuel class (gas/electric/oil) $\times$ era (Victorian/Edwardian/Interwar/post-war/modern); the council-stock reference (R) rows add representative social housing. Short names are used in all other HTE exhibits. For flats, ‘built form’ is the RdSAP heat-loss exposure category (number of party walls), not a house typology. Baseline EPC = control-pool mean EPC score of the cell (the level a treatment effect shifts from). Floor-area terciles cut at 64 and 86 sqm on the control pool. Source: `hte_archetype_definitions.LA.csv`, `hte_reference_cells.LA.csv`, `hte_cell_baselines.LA.csv` (script 06c).

Table 2: Composition vs. Treatment: Reweighted Effects, EPC Score

Treatment	ATT (SAP pts)	τ_{rw} (full)	τ_{rw} (10 arch.)	Composition	Wald p
For-profit (all)	+0.45***	+0.16 (0.050)	+0.01 (0.068)	+0.29	<0.001
UK for-profit	+0.44***	+0.15 (0.051)	+0.00 (0.069)	+0.29	<0.001
Foreign for-profit	+1.53***	+0.64 (0.138)	+0.36 (0.210)	+0.88	<0.001
Non-profit	+3.15***	+3.29 (0.119)	+3.57 (0.145)	-0.14	<0.001
Tax haven (all)	+1.50***	+0.65 (0.141)	+0.23 (0.247)	+0.85	<0.001
Public sector	+3.38***	+2.62 (0.158)	+2.41 (0.223)	+0.76	<0.001

Notes: Matched-pair design, Baseline spec, base core, LA-clustered. ATT = pooled pair-difference estimate (identical to the PSM + Subclass FE coefficient). τ_{rw} = cell CATEs reweighted to the control-pool covariate distribution (delta-method SEs in parentheses); full = all estimated cells, 10 arch. = the diverse narrative archetype set (Table 1). Composition = ATT – τ_{rw} (full): the part of the pooled effect attributable to the treated group’s covariate mix. Wald p : test that all cell CATEs are equal. Source: `hte_reweighted.LA.csv` (script 06c).

Table 3: Composition vs. Treatment: Reweighted Effects, Below EPC C

Treatment	ATT (pp)	τ_{rw} (full)	τ_{rw} (10 arch.)	Composition	Wald p
For-profit (all)	-3.12***	-2.80 (0.239)	-2.60 (0.315)	-0.32	<0.001
UK for-profit	-3.10***	-2.78 (0.240)	-2.54 (0.314)	-0.32	<0.001
Foreign for-profit	-5.38***	-2.99 (0.670)	-1.64 (1.109)	-2.39	<0.001
Non-profit	-14.76***	-14.83 (0.537)	-15.91 (0.675)	+0.08	<0.001
Tax haven (all)	-5.44***	-3.03 (0.694)	-0.67 (1.331)	-2.41	<0.001
Public sector	-16.55***	-12.58 (0.785)	-11.70 (1.018)	-3.97	<0.001

Notes: Matched-pair design, Baseline spec, base core, LA-clustered. ATT = pooled pair-difference estimate (identical to the PSM + Subclass FE coefficient). τ_{rw} = cell CATEs reweighted to the control-pool covariate distribution (delta-method SEs in parentheses); full = all estimated cells, 10 arch. = the diverse narrative archetype set (Table 1). Composition = ATT - τ_{rw} (full): the part of the pooled effect attributable to the treated group's covariate mix. Wald p : test that all cell CATEs are equal. Coefficients in percentage points. Source: `hte_reweighted_LA.csv` (script 06c).

Table 4: Typical (Modal) Property by Ownership Type

Ownership	Property type	Built form	Age band	Main fuel	Share	Mean EPC	Estimable
For-Profit	House	Mid-Terrace	1900-1929	Mains gas	2.2%	67.1	yes
UK For-Profit	House	Mid-Terrace	1900-1929	Mains gas	2.3%	66.9	yes
Foreign For-Profit	Flat	Mid-Terrace	2003-2006	Electricity	3.0%	71.6	no
Non-Profit	House	Semi-Detached	1950-1966	Mains gas	2.9%	69.9	yes
Public Sector	House	Semi-Detached	1950-1966	Mains gas	3.7%	68.7	yes
Tax Haven	Flat	Mid-Terrace	2003-2006	Electricity	2.9%	71.6	no
Tax Haven For-Profit	Flat	Mid-Terrace	2003-2006	Electricity	3.0%	71.8	no
Tax Haven Non-Profit	Flat	Mid-Terrace	1900-1929	Mains gas	4.9%	67.5	yes
Caribbean Haven	Flat	Mid-Terrace	2003-2006	Electricity	2.2%	69.1	no
British Haven	Flat	Mid-Terrace	2003-2006	Electricity	2.9%	71.9	no
European Haven	Flat	Mid-Terrace	2003-2006	Electricity	3.0%	72.3	no
Other Haven	Flat	Mid-Terrace	2003-2006	Electricity	5.1%	68.0	no
Foreign Non-Profit	Flat	Mid-Terrace	1930-1949	Mains gas	7.9%	67.7	no
UK Non-Profit	House	Semi-Detached	1950-1966	Mains gas	2.9%	69.9	yes

Notes: For each ownership type, the single most common covariate cell among its *treated* eligible properties (property type $\times$ built form $\times$ age band $\times$ main fuel $\times$ floor tercile; tercile omitted here for space). Share = share of the treated group falling in that modal cell. Mean EPC = mean current energy-efficiency score of the treated group. Estimable = the modal cell is in the archetype whitelist *and* has ≥ 50 matched pairs for that ownership type, so a matched CATE exists for it; distinctive treated cells (e.g. prime flats) often lack matched support and so are not estimable, which is why the matched archetypes reflect the common control stock instead. Source: `hte_typical_property_LA.csv` (06c).

Table 5: Archetype $\times$ Ownership Treatment Effects with Baseline Anchor (EPC Score)

A	Archetype	Baseline	For-profit	Non-profit	Public	Tax haven	Foreign FP
A1	Edwardian terrace (medium)	62.1	-0.37	+4.38	+3.45	-0.17	+0.23
A2	Edwardian flat	67.0	+0.17	+1.72	+0.90	-0.02	+1.10
A3	Post-war semi	63.5	+0.18	+3.63	+3.75	-0.71	-1.55
A4	Edwardian electric flat	56.2	+2.33	+7.12	+3.52	+7.16	+6.75
A5	1980s terrace	69.9	+0.27	+1.74	+1.22	-0.27 [†]	-0.20 [†]
A6	Victorian oil detached	42.7	-0.51	-1.30	-8.25	+0.90	-0.32
A7	Post-war bungalow	63.1	-0.17	+3.05	+3.08	—	—
A8	Edwardian terrace (large)	60.9	+0.58	+3.55	+2.64	+0.41	+0.63
A9	Interwar semi	60.6	+0.08	+4.80	+4.81	+0.47	-0.13
A10	Victorian terrace	61.1	-0.88	+3.50	+1.24	-1.17	-0.97
<i>Council-stock reference (representative social housing)</i>							
R	Council flat	66.7	+0.99	+3.49	+3.17	+1.37	+1.74
R	Council semi	63.5	+0.18	+3.63	+3.75	-0.71	-1.55

Notes: Each cell is the matched-pair CATE (treated – control) in EPC score for that ownership type, holding the property fixed at the row archetype; positive = more efficient than the private-individual control. Baseline = control-pool mean EPC of the archetype (the level the effect shifts from). Effects are estimated per owner and flagged, not dropped: [†] = suspect (thin / few-cluster support, still reported); — = no matched support. Archetypes (A) are the diverse representative control-pool cells of Table 1; the council-stock reference (R) rows anchor the public / non-profit effect on representative social housing (the post-war gas semi and a post-war purpose-built gas flat; the flat's RdSAP built form is 'semi-detached', i.e. one party wall, not a house typology). Baseline spec, base core, LA-clustered. Source: `hte_archetype_matrix.LA.csv` + `hte_cell_baselines.LA.csv` (06c).

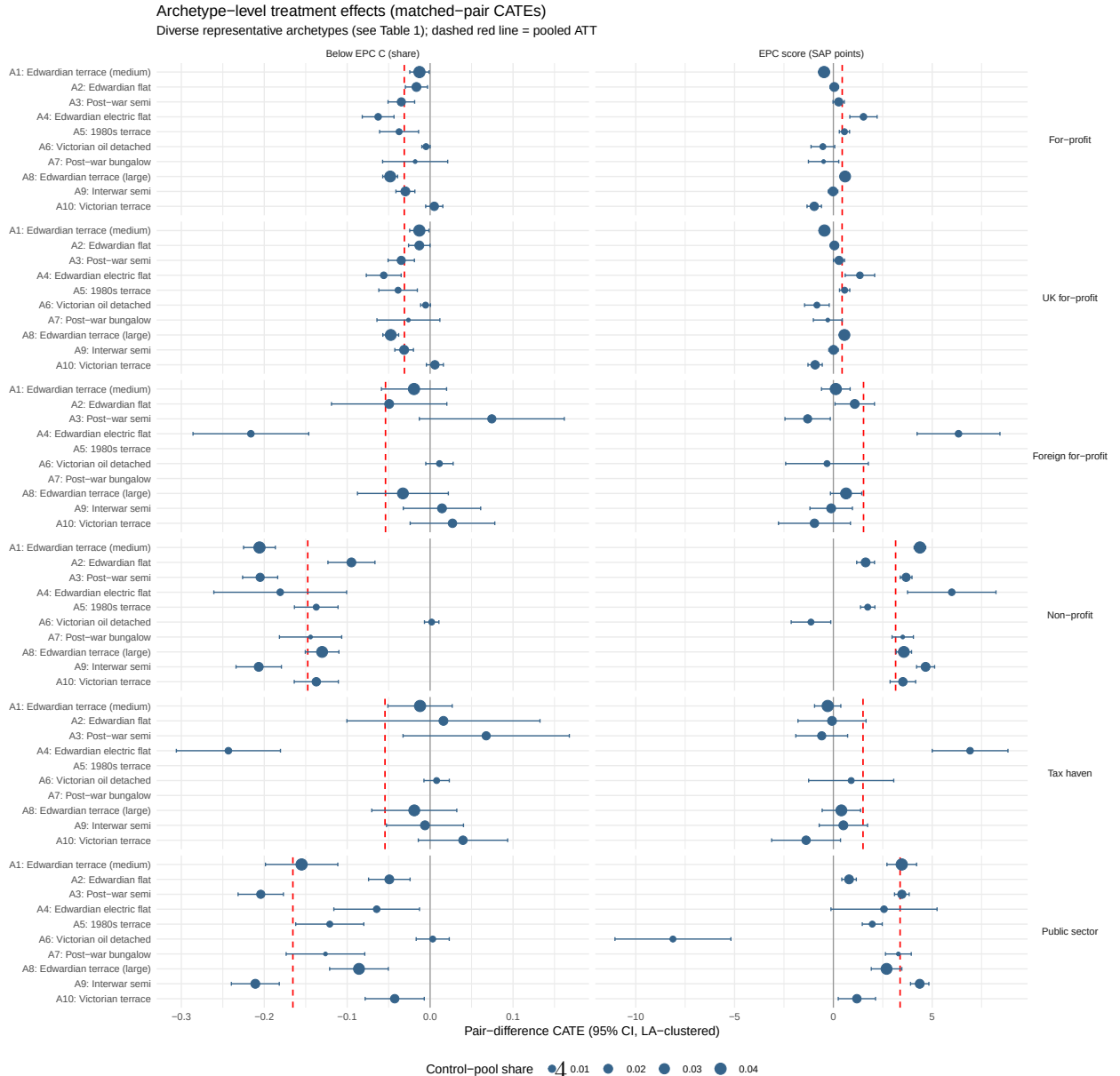


Figure 1: Archetype-level treatment effects (matched-pair CATEs) for the six headline ownership

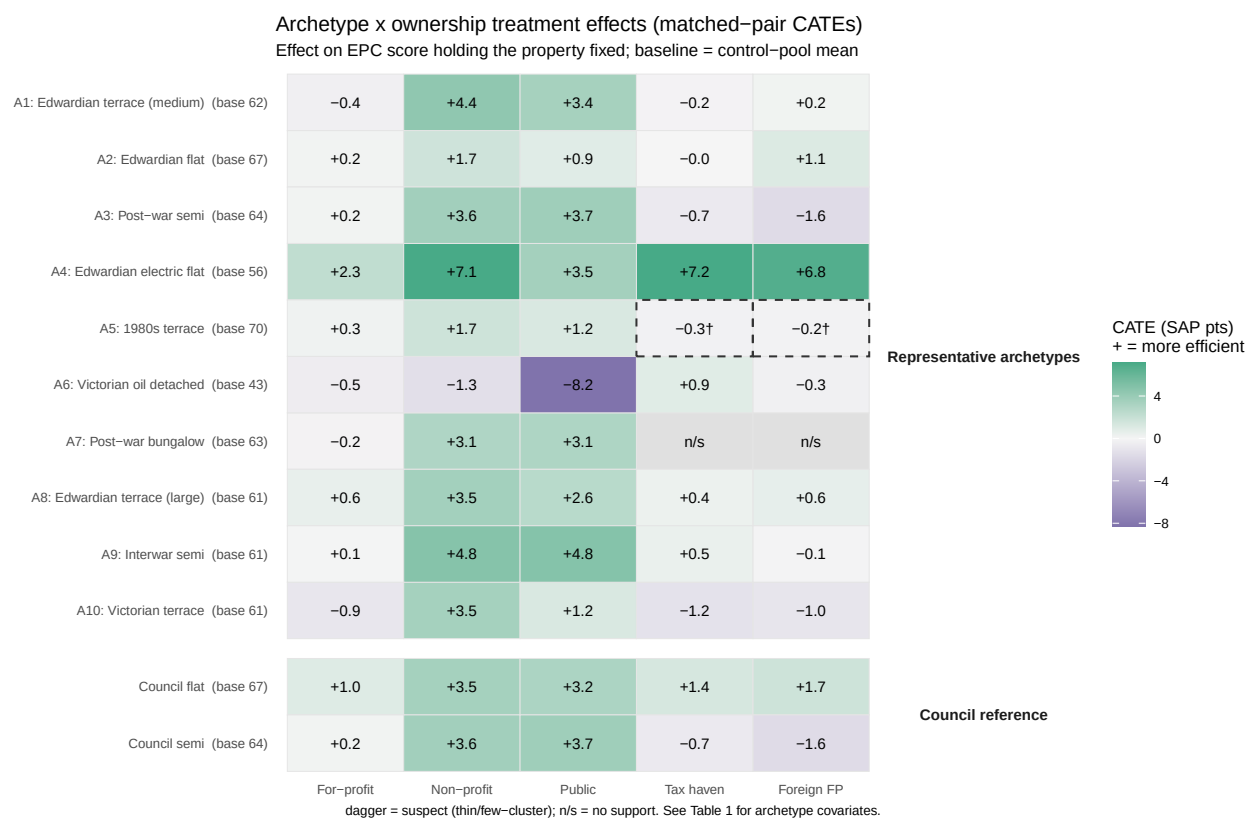


Figure 2: Archetype $\times$ ownership matrix as a heatmap (green = more efficient, purple = less). Baseline EPC in each row label; dashed outline = suspect support; grey = no matched support. Council-stock references shown as a separate panel.

2 Matching Attrition

Task 2. Property and Local-Authority counts at each stage of the matching funnel.

Table 6: Matching Attrition Funnel: Treated Units and LA Survival by Stage

Stage	For-profit	Foreign FP	Tax haven	Non-profit	Public
<i>Panel A: Baseline spec, base core</i>					
Eligible (raw, core sample)	3,027,035	174,526	174,947	2,489,856	1,435,432
Complete covariate cases	2,282,352	90,252	90,336	2,002,000	1,361,592
Passed size gate	2,282,352	90,109	90,202	2,002,000	1,361,460
Matched (1:1 PSM)	913,094	40,233	40,385	559,071	360,774
Caliper PS ≤ 0.2	23,356	34,412	34,314	37,101	26,631
Caliper PS ≤ 0.1	4,214	25,625	25,503	6,728	8,638
LAs at start	346	346	346	346	346
LAs with eligible sample	346	345	346	346	346
LAs passing gate	346	319	319	346	325
LAs with matches	346	325	324	346	331
LAs in headline regression	346	329	326	346	338
<i>Panel B: Price Paid spec, ppd core</i>					
Eligible (raw, core sample)	2,000,667	119,985	120,657	332,225	278,108
Complete covariate cases	1,638,777	65,014	65,245	261,186	261,863
Passed size gate	1,637,430	64,887	65,113	261,135	260,469
Matched (1:1 PSM)	218,576	5,423	5,290	27,607	21,243
Caliper PS ≤ 0.2	13,529	4,033	3,873	16,852	8,393
Caliper PS ≤ 0.1	3,959	3,012	2,859	9,219	4,683
LAs at start	346	346	346	346	346
LAs with eligible sample	346	343	344	345	337
LAs passing gate	344	320	317	344	295
LAs with matches	345	262	255	343	275
LAs in headline regression	345	269	261	344	278

Notes: Counts of treated properties surviving each stage of the matching funnel, pooled across local authorities, plus the number of LAs surviving each stage. Eligible = non-missing treatment indicator within the core sample. Complete cases = non-missing matching covariates. Size gate = per-LA minimum treated/control counts (sparse cores: > 5 treated, > 25 control; other cores: > 10 treated, > 50 control). Matched = 1:1 nearest-neighbour propensity-score matches with exact strata. Calipers are post-hoc propensity-score filters as in the PS ≤ 0.2 / PS ≤ 0.1 models. “LAs in headline regression” = distinct-LA cluster count of the PSM + Subclass FE model (outcome `current_energy_efficiency`). Source: `attrition_funnel_LA.csv` (script 05b).

3 Transaction-Vintage Gradient

Task 3. Does the ownership efficiency gap fade with time since the property last transacted? Matched pair-difference gradient and the unmatched/matched estimator ladder. Descriptive: selection into transaction timing is not random.

Table 7: EPC-Score Penalty by Time Since Last Transaction

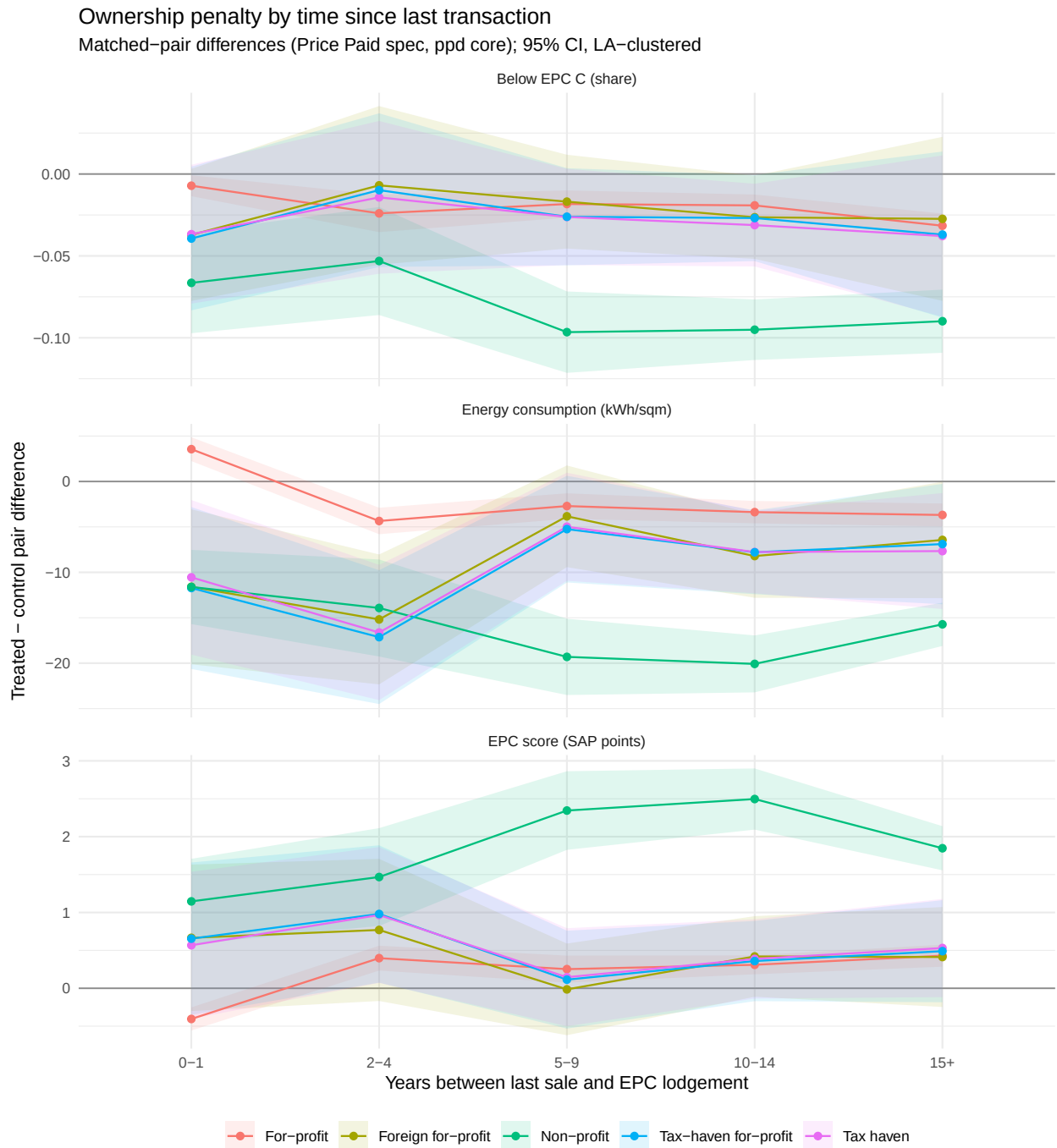
Years since sale	For-profit	Foreign FP	Tax haven	TH for-profit	Non-profit
0-1	-0.40*** (0.08)	+0.67 (0.49)	+0.57 (0.49)	+0.65 (0.51)	+1.15*** (0.29)
2-4	+0.40*** (0.08)	+0.77 (0.48)	+0.97* (0.46)	+0.98* (0.46)	+1.47*** (0.33)
5-9	+0.25** (0.09)	-0.01 (0.31)	+0.15 (0.33)	+0.11 (0.33)	+2.34*** (0.26)
10-14	+0.31*** (0.06)	+0.42 (0.27)	+0.39 (0.27)	+0.36 (0.27)	+2.50*** (0.21)
15+	+0.43*** (0.07)	+0.41 (0.34)	+0.53 (0.33)	+0.49 (0.34)	+1.85*** (0.15)

Notes: Mean matched-pair difference in EPC score (treated – control) by years between the property’s most recent sale and its EPC lodgement. Price Paid specification, ppd matching core: pairs exact-match on both `lodgement_year` and `ppd_year_transfer`, so the vintage is constant within pair. LA-clustered SEs in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. **Descriptive:** selection into transaction timing is not random; this is a gradient shape, not identification of retrofit dynamics. Sales postdating the EPC are excluded (share reported in `txn_vintage_gradient_LA.csv`). Source: script 06d.

Table 8: Estimator Ladder: EPC-Score Ownership Penalty by Transaction Vintage

Ownership	Years since sale	Simple (raw)	Simple (adj.)	Matched
For-profit	0-1	+4.06	+0.66	-0.40
	2-4	+5.44	+1.26	+0.40
	5-9	+5.45	+1.09	+0.25
	10-14	+5.15	+0.63	+0.31
	15+	+3.57	+0.65	+0.43
Foreign for-profit	0-1	+9.90	+3.02	+0.67
	2-4	+9.96	+2.64	+0.77
	5-9	+10.17	+2.56	-0.01
	10-14	+9.97	+2.43	+0.42
	15+	+6.58	+2.40	+0.41
Tax haven	0-1	+9.94	+3.00	+0.57
	2-4	+9.95	+2.62	+0.97
	5-9	+10.26	+2.60	+0.15
	10-14	+9.95	+2.39	+0.39
	15+	+6.66	+2.46	+0.53
Tax-haven for-profit	0-1	+10.01	+3.03	+0.65
	2-4	+10.01	+2.63	+0.98
	5-9	+10.29	+2.62	+0.11
	10-14	+10.02	+2.42	+0.36
	15+	+6.74	+2.47	+0.49
Non-profit	0-1	+6.88	+2.10	+1.15
	2-4	+6.79	+2.19	+1.47
	5-9	+6.16	+2.18	+2.34
	10-14	+5.16	+1.97	+2.50
	15+	+4.05	+1.81	+1.85

Notes: Ownership penalty in EPC score (treated – control) by years between the property’s most recent sale and its EPC lodgement. *Simple (raw)*: unmatched per-bin gap from $y \sim i(\text{bin}, \text{owner}) + \text{bin}$. *Simple (adjusted)*: same, plus property-type, built-form, age-band, main-fuel fixed effects and floor area / habitable rooms (unmatched but composition-adjusted). *Matched*: mean pair difference (Price Paid spec, ppd core; exact-matched on vintage). Raw includes composition by vintage; matched strips it; adjusted sits between. LA-clustered SEs. **Descriptive:** selection into transaction timing is not random. Source: script 06d.



Descriptive gradient: selection into transaction timing is not random; not identification of retrofit dynamics. Sales postdating the EPC excluded.

Figure 3: Matched-pair ownership penalty by years since last sale (levels; 95% CI ribbons; panel per outcome).

Ownership penalty by time since last transaction: estimator ladder

Unmatched raw and covariate-adjusted vs matched pair-difference (ppd core); 95% CI, LA-clustered; free y-scale per panel

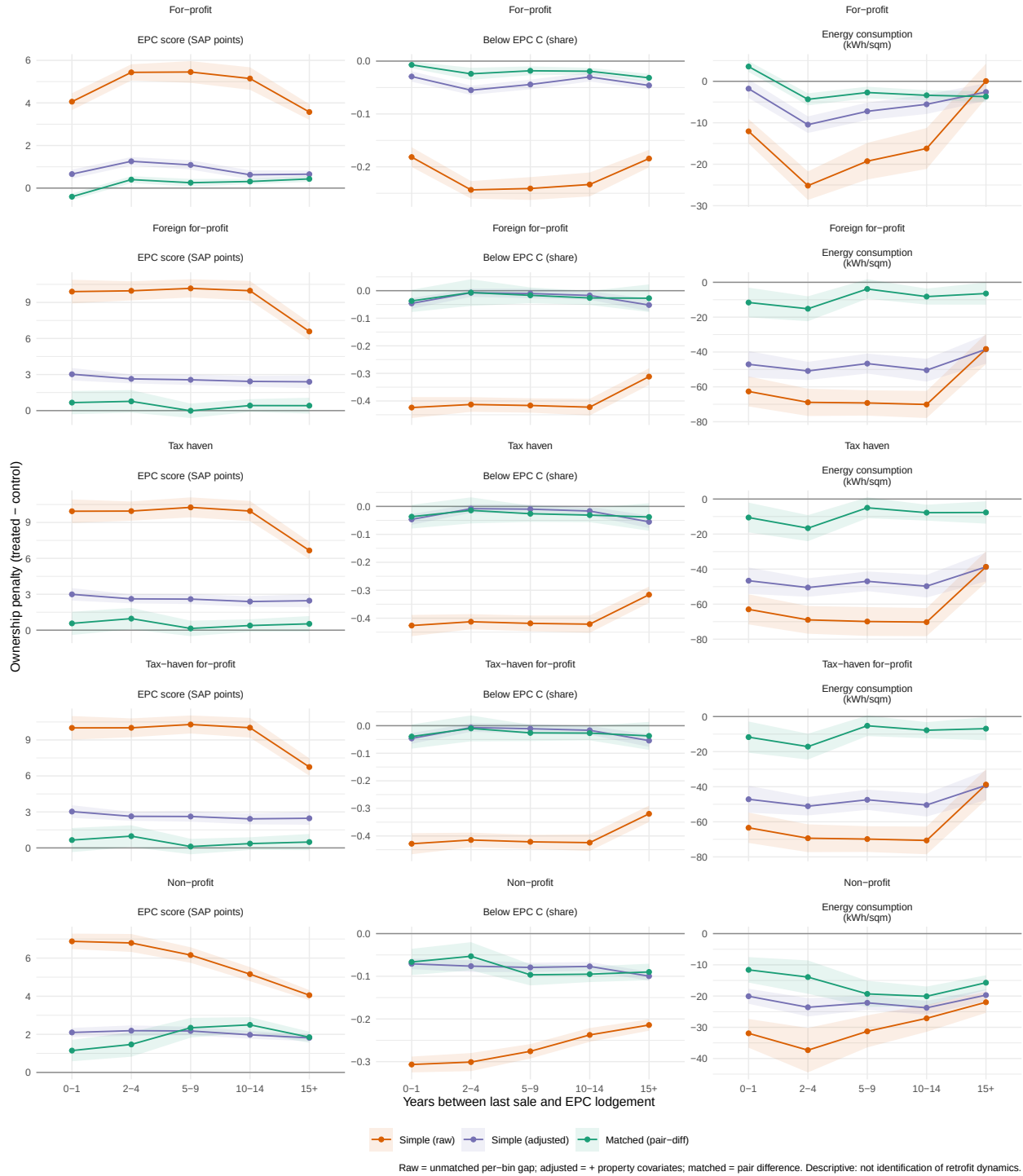


Figure 4: Estimator ladder: unmatched raw and covariate-adjusted vs. matched pair-difference penalty, free y -scale per panel.

4 Foreign Non-Profits and Tax Havens

Task 4. Descriptive universe of offshore and foreign non-profit ownership (company-level public register data only).

Table 9: Tax-Haven Owned Rental Properties by Jurisdiction of Incorporation

Jurisdiction	Properties	Titles	Proprietors	Mean EPC	Share <C
GUERNSEY	76,691	5,298	784	78.2	10.8%
JERSEY	56,714	5,036	1,310	77.2	14.5%
BRITISH VIRGIN ISLANDS	19,746	6,811	3,825	72.4	28.1%
ISLE OF MAN	11,790	3,249	1,192	71.7	30.3%
GIBRALTAR	1,999	737	278	66.5	48.9%
HONG KONG	983	566	313	74.3	20.5%
CYPRUS	872	436	243	72.8	30.4%
PANAMA	735	375	278	66.5	48.2%
BAHAMAS	707	300	189	69.9	36.5%
SEYCHELLES	633	446	314	67.9	43.0%
CAYMAN ISLANDS	489	206	124	72.9	27.8%
BERMUDA	378	76	42	58.9	90.7%
BELIZE	295	158	86	73.5	27.1%
MAURITIUS	272	220	149	68.5	37.9%
LIECHTENSTEIN	239	136	75	63.0	54.8%

Notes: Properties in the tax-haven ownership group (IMF tax-haven jurisdiction lists applied to CCOD/OCOD country of incorporation), deduplicated by UPRN. Top 15 jurisdictions by property count. Share <C = share of properties below EPC band C (SAP < 69). Source: `universe_by-jurisdiction.csv` (script 08b).

Table 10: Largest Tax-Haven and Foreign Non-Profit Proprietors (Public Register)

Group	Proprietor (registered company)	Properties	LAs	Mean EPC	Share <C
Tax haven (all)	ABACUS LAND 4 LIMITED (GUERNSEY)	18,870	240	76.2	12.9%
Tax haven (all)	ADRIATIC LAND 3 LIMITED (GUERNSEY)	14,490	204	78.8	7.9%
Tax haven (all)	ABACUS LAND 1 (HOLDCO 1) LIMITED (JERSEY)	12,613	180	76.8	11.3%
Tax haven (all)	PENNINE TRUSTEES LIMITED (JERSEY)	11,216	48	79.2	6.4%
Tax haven (all)	ADRIATIC LAND 11 LIMITED (GUERNSEY)	8,544	152	78.7	9.3%
Tax haven (all)	ADRIATIC LAND 5 LIMITED (GUERNSEY)	6,625	112	78.9	12.3%
Tax haven (all)	ADRIATIC LAND 12 LIMITED (GUERNSEY)	3,624	54	82.2	3.6%
Tax haven (all)	ADRIATIC LAND 9 LIMITED (GUERNSEY)	3,189	72	81.6	3.5%
Tax haven (all)	ADRIATIC LAND 10 LIMITED (GUERNSEY)	3,038	53	81.9	8.3%
Tax haven (all)	ADRIATIC LAND 7 LIMITED (GUERNSEY)	2,825	78	80.9	5.0%
Foreign non-profit	THE DOLPHIN SQUARE ESTATE S.A.R.L. (LUXEMBOURG)	1,054	1	70.3	20.2%
Foreign non-profit	BALTIMORE WHARF SLP (JERSEY)	937	1	82.0	0.4%
Foreign non-profit	FREEHOLD REVERSIONS PARTNERSHIP INCORPORATED (GUERNSEY)	503	15	79.2	3.9%
Foreign non-profit	WHITEWOOD (RING) UK S.A.R.L. (LUXEMBOURG)	421	1	85.7	0.0%
Foreign non-profit	GREENGATE S.A R.L. (LUXEMBOURG)	351	1	82.6	0.3%

Notes: Proprietor names are registered company names from the public CCOD/OCOD registers (company-level data only; no individual beneficial owners). Properties = distinct UPRNs held under the (normalised) proprietor name. Source: `top_proprietors.csv` (script 08b).